# Ruiwen WANG

AI-infrastructure & ML-systems engineer — large-scale LLM pretraining (5+ years at Huawei Paris).
I make LLM training faster and more efficient — performance modelling, bottleneck analysis,
and automatic parallelization at production scale; open to **AI-infra / ML-systems roles**.

## Contact


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French (EU) · open to relocation


## Highlights

**16,384 NPUs** production scale
up to **1.7**$\times$ faster training
dense · MoE · multimodal models, up to **1T+** parameters
**6** publications · **3** patent families

## Core Areas

**Training:** large-scale LLM pretraining · MoE · long context

**Parallelism:** DP / TP / PP / EP · ZeRO-style sharding · pipeline scheduling · recomputation

**Planning:** cost models (compute, memory, communication) · ILP / CP optimisation · graph (IR) level

**Performance:** profiling · bottleneck analysis · tuning

## Awards

**Best Poster Award** — Euro-Par 2025

**Golden Prize** — Huawei Challenge for Automatic Load Balancing, 2024

**Letter of Appreciation** — China Telecom, 2025

## Languages

| | |
|---|---|
| Mandarin (native) | ●●●●●● |
| Cantonese (native) | ●●●●●● |
| French (fluent) | ●●●●●○ |
| English (fluent) | ●●●●●○ |

## Programming

Python, C++, Java

**Frameworks:** MindSpore, MindFormers, PyTorch, Megatron-LM

**Stack:** Ascend CANN, HCCL (NCCL-style collectives)

**Tools:** Linux, Git, ILP / CP solvers

## Online


ruiwen.wang
linkedin.com/in/ruiwen-wang-ai
Google Scholar


## Experience

**Ph.D. Researcher (CIFRE) — LLM Training Systems** 2023 – 2026
Huawei Paris Research Center · Sorbonne Université & EURECOM · Paris

- Built compute, memory, and communication cost models and optimisation-based planners for Huawei's LLM-pretraining stack — shipped in MindSpore / MindFormers, benchmarked against Megatron-LM.
- Took planners from research prototype to production: configuration guidance, debuggable traces, and regression benchmarking across model families and cluster scales.

**Research Apprentice → Research Engineer — AI Infrastructure** 2021 – 2023
Huawei Paris Research Center · Distributed & Parallel Computing Lab · Paris

- Profiling, bottleneck analysis, and performance tuning for large-scale LLM training and inference on Ascend-910 production clusters.

## Deployments & Impact

- **Scale:** Ascend-910 clusters up to **16,384 NPUs** — dense, MoE, and multimodal models up to **1T+** parameters (DeepSeek, Llama, Mistral / Mixtral, Qwen).
- **Production training:** supported China Telecom's DeepSeek-MoE deployment (Letter of Appreciation) and TeleChat3-MoE training.
- **Portability:** cost surfaces separate model structure from hardware rates — plans retarget across clusters and accelerator families without re-profiling.

## Selected Systems

- **PRISM** — profiling-free, layer-level symbolic cost model; predicts memory (**92–96%** accuracy), communication, and compute time *before* a run.
- **BMPipe** — co-optimises pipeline bubbles and stage memory; up to **1.7**$\times$ faster at **16,384 NPUs**.
- **$H^2O$** — searches all parallelism dimensions jointly; **+37%** throughput over the automated baseline, **+22%** over expert-tuned plans.
- **Concord** — re-plans only what a model edit changes; $\sim$**2.7**$\times$ faster re-planning, provably identical plans.
- **ManuMatic** — operator-level strategy injection: pin a few performance-critical operators, automate the rest; up to **2.4**$\times$ over the baseline plan.

## Publications

- **PRISM: Profiling-Free Symbolic Memory-Driven Strategy Planner.** SCA/HPC Asia 2026.
- **ManuMatic: Strategy Injection for Robust Automatic Hybrid Parallelism.** IFIP NPC 2025.
- **BMPipe: Bubble–Memory Co-optimization Strategy Planner.** IEEE CLUSTER 2025.
- **$H^2O$: Holistic Hyper-Parameter Optimization.** Euro-Par 2025 — Best Poster.
- Co-author: **MLLM Pipeline Bubble Modeling** (SCA/HPC Asia 2026); **TeleChat3-MoE** report (arXiv 2025).
- **Concord: Dependency-Keyed Per-Layer Primitives for Re-Planning** — under submission.

## Patents

- Chong Li, Pierre Leca, Thibaut Tachon, **Ruiwen Wang**, Haoran Wang. *Devices and methods for generating execution plans for a machine learning model.* WO 2025/020165 A1 (PCT), Huawei — published 2025.
- Chong Li, **Ruiwen Wang**, Haoran Wang, Fan Yu. *A load-balancing method to improve large-model performance.* Huawei — filed (pending).
- Chong Li, Thibaut Tachon, **Ruiwen Wang**, Haoran Wang. *A multi-dimension parallelism configuration method for large language model training.* Huawei — filed (pending).

## Education

**Ph.D. Computer Science (CIFRE)** — Sorbonne Université & EURECOM 2023 – 2026
Advisors: Dr. Raja Appuswamy (EURECOM) · Dr. Chong Li (Huawei Paris).

**M.Sc. Computer Science** — Sorbonne Université 2019 – 2022
Advisors: Prof. Emmanuel Chailloux & Prof. Jacques Malenfant (both LIP6).

**B.Sc. Computer Science** — Université Paris-Saclay 2016 – 2019
Mentors: Prof. Isabelle Guyon (Google DeepMind) · Dr. Viviane Pons · Dr. Zhengying Liu (Moonshot AI).